RADIATIVE RECOMBINATION OF HYDROGENIC IONS

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Summary

Using the first three terms in the asymptotic expansion of the Kramers-Gaunt factor, calculations are made for the rate of recombination and for the mean kinetic energy of the recombining electrons.

For the study of the recombination spectra of gaseous nebulae one requires the rate coefficients for radiative recombination to individual quantum states of hydrogenic ions and for the study of the ionization equilibria one requires the radiative recombination rates summed over all quantum states (1). Further, for the study of the thermal balance in nebulae, it is necessary to calculate the rate of loss of electron kinetic energy due to recombination processes (2, 3). For hydrogenic ions all these quantities may be calculated exactly but the calculations become increasingly difficult with increasing n, where n is the principal quantum number. If errors as great as 20 per cent are of no importance one may put the Kramers–Gaunt g-factor equal to unity and hence obtain relatively simple expressions. A considerable improvement is obtained on using the asymptotic expansion of the g-factor, as derived by Menzel and Pekeris (4) and corrected by Burgess (5). From the expansion of g we obtain asymptotic expansions which enable the rate coefficients to be calculated with errors which will not exceed 2 per cent for temperatures of order 10^4 °K or less but which may be greater for temperatures of order 10^6 °K. We present a systematic tabulation of the various functions which occur in these asymptotic expansions.

1. *Coefficients for recombination to level n.*—Let $\alpha_n(Z, T)$ be the rate coefficient for radiative recombination to level n of a hydrogenic ion, Z being the nuclear charge and T the kinetic temperature; the number of recombinations on n per unit volume per unit time is then $\alpha_n N_e N_+$. The recombination coefficient is given by

$$\alpha_n(Z, T) = \frac{1}{c^2}\left(\frac{2}{\pi}\right)^{1/2} (mkT)^{-3/2}\, 2n^2 e^{I_n/kT} \int_{I_n}^{\infty} (h\nu)^2 e^{-h\nu/kT} a_n(Z, h\nu)\, d(h\nu). \qquad (1)$$

where

$$I_n = \frac{hRcZ^2}{n^2} \qquad (2)$$

is the threshold ionization energy for level n and R is the Rydberg wave number. The cross-section for photo-ionization from level n is

$$a_n(Z, h\nu) = \frac{2^6 \alpha \pi a_0^2}{3\sqrt{3}} \frac{n}{Z^2} (1 + n^2\epsilon)^{-3} g_{\text{II}}(n, \epsilon) \qquad (3)$$

where α is the fine-structure constant, a_0 the Bohr radius and

$$h\nu = hRcZ^2\left(\frac{1}{n^2} + \epsilon\right). \qquad (4)$$

Note that the energy of the ejected electron is $Z^2\epsilon$ in Rydberg units (13·60 eV). The Kramers–Gaunt factor g_{II} in (3) is of order unity.

From equations (1) to (4) we obtain

$$\alpha_n(Z, T) = \mathscr{D} Z \frac{\lambda^{1/2}}{n} x_n S_n(\lambda) \tag{5}$$

where

$$\mathscr{D} = \frac{2^6}{3}\left(\frac{\pi}{3}\right)^{1/2} \alpha^4 c a_0^2$$

$$= 5{\cdot}197 \times 10^{-14}\ \mathrm{cm^3\ sec^{-1}}, \tag{6}$$

$$\lambda = \frac{hRcZ^2}{kT}$$

$$= 157890 \frac{Z^2}{T} \qquad (T \text{ in } {}^\circ\mathrm{K}), \tag{7}$$

$$x_n = \lambda / n^2 \tag{8}$$

and

$$S_n(\lambda) = \int_0^\infty \frac{g_{\mathrm{II}}(n, \epsilon) \mathrm{e}^{-x_n u}}{(1+u)}\, du \tag{9}$$

with $u = n^2\epsilon$.

The asymptotic expansion for g_{II} is (4, 5)

$$g_{\mathrm{II}}(n, \epsilon) = 1 + 0{\cdot}1728 n^{-2/3}(u+1)^{-2/3}(u-1) - 0{\cdot}0496 n^{-4/3}(u+1)^{-4/3}(u^2 + \tfrac{4}{3}u + 1) + \ldots. \tag{10}$$

Substitution of (10) in (9) gives

$$S_n(\lambda) = S^{(0)}(x_n) + \lambda^{-1/3} S^{(1)}(x_n) + \lambda^{-2/3} S^{(2)}(x_n) + \ldots \tag{11}$$

where

$$S^{(0)}(x) = \mathrm{e}^x \mathscr{E}i(x), \tag{12}$$

$$S^{(1)}(x) = 0{\cdot}1728 x^{1/3} \mathrm{X}_1(x), \tag{13}$$

$$S^{(2)}(x) = -0{\cdot}0496 x^{2/3} \mathrm{X}_2(x), \tag{14}$$

and

$$\mathscr{E}i(x) = \int_x^\infty \frac{\mathrm{e}^{-v}}{v}\, dv, \tag{15}$$

$$\mathrm{X}_1(x) = \int_0^\infty (u+1)^{-5/3}(u-1) \mathrm{e}^{-xu}\, du, \tag{16}$$

$$\mathrm{X}_2(x) = \int_0^\infty (u+1)^{-7/3}(u^2 + \tfrac{4}{3}u + 1) \mathrm{e}^{-xu}\, du. \tag{17}$$

Extensive tables of the exponential integral, $\mathscr{E}i(x)$, are available (6). For all finite x

$$S^{(0)}(x) = \mathrm{e}^x \left\{ -\ln x - \gamma + \frac{x}{1.1!} - \frac{x^2}{2.2!} + \frac{x^3}{3 \cdot 3!} - \ldots \right\} \tag{18}$$

where $\gamma = 0{\cdot}5772\ldots$ is the Euler–Mascheroni constant. In the limit of large x, $S^{(0)}$ has asymptotic form

$$S^{(0)}(x) \underset{x\to\infty}{\sim} \frac{1}{x}\left\{1 - \frac{1!}{x} + \frac{2!}{x^2} - \frac{3!}{x^3} + \ldots\right\}. \tag{19}$$

TABLE I

x	$xS^{(0)}$	$xS^{(1)}$	$xS^{(2)}$	x	$xS^{(0)}$	$xS^{(1)}$	$xS^{(2)}$	x	$xS^{(0)}$	$xS^{(1)}$	$xS^{(2)}$
0·020	·06845	·00417	−·00120	0·20	·2987	·0074	−·0095	2·0	·7226	−·0882	−·0625
·022	·07335	·00444	−·00131	·22	·3140	·0068	−·0103	2·2	·7384	−·0972	−·0674
·024	·07808	·00469	−·00142	·24	·3284	·0060	−·0110	2·4	·7524	−·1058	−·0722
·026	·08268	·00493	−·00153	·26	·3419	·0053	−·0118	2·6	·7649	−·1141	−·0768
·028	·08714	·00516	−·00164	·28	·3547	·0044	−·0126	2·8	·7761	−·1220	−·0814
·030	·09148	·00538	−·00175	·30	·3668	·0035	−·0133	3·0	·7862	−·1295	−·0859
·032	·09570	·00558	−·00186	·32	·3783	·0025	−·0141	3·2	·7955	−·1368	−·0904
·034	·09982	·00578	−·00197	·34	·3892	·0016	−·0148	3·4	·8039	−·1438	−·0947
·036	·10385	·00597	−·00207	·36	·3996	+·0005	−·0155	3·6	·8117	−·1506	−·0989
·038	·10778	·00614	−·00218	·38	·4096	−·0005	−·0162	3·8	·8188	−·1571	−·1031
·040	·11163	·00631	−·00228	·40	·4191	−·0016	−·0170	4·0	·8254	−·1634	−·1072
·045	·1209	·0067	−·0025	·45	·4413	−·0044	−·0187	4·5	·8399	−·1784	−·1174
·050	·1297	·0070	−·0028	·50	·4615	−·0072	−·0204	5·0	·8521	−·1923	−·1272
·055	·1382	·0073	−·0030	·55	·4799	−·0101	−·0221	5·5	·8626	−·2053	−·1367
·060	·1462	·0076	−·0033	·60	·4968	−·0130	−·0237	6·0	·8716	−·2174	−·1460
·065	·1540	·0078	−·0035	·65	·5124	−·0160	−·0253	6·5	·8795	−·2289	−·1550
·070	·1615	·0080	−·0038	·70	·5269	−·0190	−·0269	7·0	·8865	−·2397	−·1638
·075	·1687	·0082	−·0040	·75	·5404	−·0220	−·0284	7·5	·8927	−·2499	−·1723
·080	·1757	·0083	−·0043	·80	·5530	−·0250	−·0299	8·0	·8982	−·2596	−·1807
·085	·1824	·0085	−·0045	·85	·5648	−·0279	−·0314	8·5	·9032	−·2689	−·1890
·090	·1889	·0086	−·0047	·90	·5759	−·0308	−·0329	9·0	·9077	−·2777	−·1970
·095	·1953	·0087	−·0050	·95	·5864	−·0337	−·0344	9·5	·9118	−·2862	−·2049
·100	·2015	·0087	−·0052	1·00	·5963	−·0366	−·0359	10·0	·9156	−·2944	−·2127
·11	·2133	·0088	−·0056	1·1	·6146	−·0422	−·0388	11	·9223	−·3099	−·228
·12	·2245	·0088	−·0061	1·2	·6311	−·0478	−·0416	12	·9279	−·3242	−·243
·13	·2352	·0088	−·0065	1·3	·6461	−·0532	−·0444	13	·9328	−·3376	−·257
·14	·2454	·0087	−·0070	1·4	·6598	−·0586	−·0471	14	·9370	−·3502	−·272
·15	·2552	·0086	−·0074	1·5	·6724	−·0638	−·0497	15	·9408	−·3622	−·285
·16	·2646	·0084	−·0078	1·6	·6840	−·0689	−·0523	16	·9441	−·3736	−·299
·17	·2736	·0082	−·0082	1·7	·6947	−·0739	−·0549	17	·9471	−·3844	−·312
·18	·2823	·0080	−·0086	1·8	·7047	−·0788	−·0575	18	·9498	−·3948	−·325
·19	·2906	·0077	−·0091	1·9	·7140	−·0835	−·0600	19	·9522	−·4047	−·337
·20	·2987	·0074	−·0095	2·0	·7226	−·0882	−·0625	20	·9544	−·4147	−·350

The integrals X_1 and X_2 may be expressed in terms of confluent hypergeometric functions (7). We obtain

$$X_1(x) = -3 + (1+3x)\Psi(1, \tfrac{4}{3}; x) \tag{20}$$

and

$$X_2(x) = -\tfrac{3}{2}(1+x) + (1+2x+\tfrac{3}{2}x^2)\Psi(1, \tfrac{5}{3}; x) \tag{21}$$

where

$$(c-1)\Psi(1,c;x) = \Gamma(c)x^{1-c}e^x - \Phi(1,c;x) \tag{22}$$

and where the series

$$\Phi(1,c;x) = 1 + \frac{x}{c} + \frac{x^2}{c(c+1)} + \dots \tag{23}$$

converges for all finite x. In the limit of large x,

$$\Psi(1,c;x) \underset{x\to\infty}{\sim} \left\{1 - \frac{(2-c)}{x} + \frac{(2-c)(3-c)}{x^2} - \dots\right\}. \tag{24}$$

The functions X_1 and X_2 have been evaluated using (23) for $x \leqslant 2$, numerical integration for $2 \leqslant x \leqslant 10$ and the asymptotic expansion (24) for $x \geqslant 10$. Table I gives $xS^{(0)}$, $xS^{(1)}$ and $xS^{(2)}$ for $0{\cdot}02 \leqslant x \leqslant 20$. The intervals in x are such that linear interpolation will introduce errors no greater than 2 in the last significant figure. To the accuracy of Table I, for small x,

$$\left.\begin{aligned} xS^{(0)} &= x\,e^x\{-\ln x - 0{\cdot}5772 + x\} \\ xS^{(1)} &= 0{\cdot}4629x(1+4x) - 1{\cdot}0368x^{4/3}\left(1+\frac{15}{8}x\right) \\ xS^{(2)} &= -0{\cdot}0672x(1+3x) + 0{\cdot}1488x^{5/3}\left(1+\frac{9}{5}x\right) \end{aligned}\right\} x \leqslant 0{\cdot}02 \tag{25}$$

and for large x,

$$\left.\begin{aligned} xS^{(0)} &= 1 - \frac{1!}{x} + \frac{2!}{x^2} - \frac{3!}{x^3} + \frac{4!}{x^4}, \\ xS^{(1)} &= -0{\cdot}1728x^{1/3}\left\{1 - \frac{8}{(3x)} + \frac{70}{(3x)^2} - \frac{800}{(3x)^3} + \frac{11440}{(3x)^4}\right\}, \\ xS^{(2)} &= -0{\cdot}0496x^{2/3}\left\{1 - \frac{3}{(3x)} + \frac{32}{(3x)^2} - \frac{448}{(3x)^3}\right\} \end{aligned}\right\} x \geqslant 20. \tag{26}$$

2. *Total recombination coefficients.*—The total rate coefficient for radiative recombination is

$$\alpha_{\text{tot}} = \sum_{n'=1}^{\infty} \alpha_{n'}. \tag{27}$$

For large n' the sum may be replaced by an integral. From

$$\sum_{n'=n}^{\infty} \alpha_{n'} \simeq \tfrac{1}{2}\alpha_n + \int_n^{\infty} \alpha_{n'}\,dn' \tag{28}$$

we obtain

$$\sum_{n'=n}^{\infty} \alpha_{n'} \simeq \tfrac{1}{2}\mathscr{D}Z\lambda^{1/2}\left\{\frac{x_n}{n}S_n(\lambda) + \sigma_n(\lambda)\right\} \tag{29}$$

where

$$\sigma_n(\lambda) = \int_0^{x_n} S_{n'}(\lambda)\,dx_{n'} \tag{30}$$

and, using (11),

$$\sigma_n(\lambda) = \sigma^{(0)}(x_n) + \lambda^{-1/3}\sigma^{(1)}(x_n) + \lambda^{-2/3}\sigma^{(2)}(x_n) + \dots \tag{31}$$

TABLE II

x	$\sigma^{(0)}$	$\sigma^{(1)}$	$\sigma^{(2)}$	x	$\sigma^{(0)}$	$\sigma^{(1)}$	$\sigma^{(2)}$	x	$\sigma^{(0)}$	$\sigma^{(1)}$	$\sigma^{(2)}$
0·020	·0877	·0053	−·0012	0·20	·4611	·0218	−·0106	2·0	1·632	−·032	−·076
·022	·0944	·0057	−·0014	·22	·4903	·0225	−·0115	2·2	1·701	−·041	−·082
·024	·1010	·0061	−·0015	·24	·5182	·0231	−·0125	2·4	1·766	−·050	−·088
·026	·1074	·0065	−·0016	·26	·5451	·0235	−·0134	2·6	1·827	−·059	−·094
·028	·1137	·0069	−·0017	·28	·5709	·0239	−·0143	2·8	1·884	−·068	−·100
·030	·1199	·0073	−·0018	·30	·5958	·0242	−·0152	3·0	1·938	−·076	−·106
·032	·1259	·0076	−·0020	·32	·6198	·0244	−·0161	3·2	1·989	−·085	−·112
·034	·1318	·0080	−·0021	·34	·6431	·0245	−·0169	3·4	2·037	−·093	−·117
·036	·1376	·0083	−·0022	·36	·6656	·0245	−·0178	3·6	2·084	−·102	−·123
·038	·1434	·0086	−·0023	·38	·6875	·0245	−·0187	3·8	2·128	−·110	−·128
·040	·1490	·0089	−·0024	·40	·7088	·0245	−·0195	4·0	2·170	−·118	−·134
·045	·1627	·0097	−·0027	·45	·7595	·0241	−·0216	4·5	2·268	−·138	−·147
·050	·1759	·0104	−·0030	·50	·8070	·0235	−·0237	5·0	2·357	−·158	−·160
·055	·1886	·0111	−·0033	·55	·8519	·0227	−·0257	5·5	2·439	−·177	−·172
·060	·2010	·0118	−·0035	·60	·8944	·0217	−·0277	6·0	2·514	−·195	−·185
·065	·2130	·0124	−·0038	·65	·9348	·0206	−·0296	6·5	2·584	−·213	−·197
·070	·2247	·0130	−·0041	·70	·9733	·0193	−·0316	7·0	2·650	−·230	−·208
·075	·2361	·0135	−·0044	·75	1·0101	·0179	−·0335	7·5	2·711	−·247	−·220
·080	·2472	·0141	−·0046	·80	1·0454	·0164	−·0354	8·0	2·769	−·264	−·231
·085	·2581	·0146	−·0049	·85	1·0792	·0148	−·0372	8·5	2·824	−·280	−·243
·090	·2687	·0150	−·0051	·90	1·1119	·0131	−·0391	9·0	2·875	−·295	−·254
·095	·2791	·0155	−·0054	·95	1·1433	·0113	−·0409	9·5	2·924	−·311	−·268
·100	·2892	·0160	−·0057	1·00	1·1736	·0095	−·0427	10·0	2·971	−·326	−·275
·11	·3090	·0168	−·0062	1·1	1·231	·006	−·046	11	3·059	−·354	−·296
·12	·3281	·0176	−·0067	1·2	1·285	+·002	−·050	12	3·139	−·382	−·317
·13	·3464	·0183	−·0072	1·3	1·337	−·002	−·053	13	3·214	−·408	−·337
·14	·3642	·0189	−·0077	1·4	1·385	−·006	−·057	14	3·283	−·434	−·356
·15	·3815	·0195	−·0082	1·5	1·431	−·011	−·060	15	3·348	−·459	−·375
·16	·3983	·0201	−·0087	1·6	1·475	−·015	−·063	16	3·409	−·482	−·394
·17	·4146	·0206	−·0092	1·7	1·517	−·019	−·067	17	3·466	−·505	−·413
·18	·4305	·0210	−·0097	1·8	1·557	−·024	−·070	18	3·520	−·528	−·431
·19	·4460	·0214	−·0101	1·9	1·595	−·028	−·073	19	3·572	−·549	−·449
·20	·4611	·0218	−·0106	2·0	1·632	−·032	−·076	20	3·621	−·570	−·467

where

$$\sigma^{(p)}(x) = \int_0^x S^{(p)}(x')\,dx'. \tag{32}$$

Using (18) it may be shown that

$$\sigma^{(0)}(x) = S^{(0)}(x) + \ln x + \gamma. \tag{33}$$

The functions $\sigma^{(0)}$, $\sigma^{(1)}$ and $\sigma^{(2)}$ are given in Table II for $0{\cdot}02 \leqslant x \leqslant 20$. For small x,

$$\left.\begin{aligned}
\sigma^{(0)} &= x\left\{\left(1+\frac{1}{2}x\right)(-\ln x - 0{\cdot}5772) + \left(1+\frac{3}{4}x\right)\right\},\\
\sigma^{(1)} &= x\left\{0{\cdot}4629(1+2x) - 0{\cdot}7776x^{1/3}\left(1+\frac{15}{14}x\right)\right\},\\
\sigma^{(2)} &= x\left\{-0{\cdot}0672\left(1+\frac{3}{2}x\right) + 0{\cdot}0893x^{2/3}\left(1+\frac{9}{8}x\right)\right\}
\end{aligned}\right\}\ x \leqslant 0{\cdot}02 \tag{34}$$

and for large x,

$$\left.\begin{aligned}
\sigma^{(0)} &= S^{(0)} + \ln x + 0{\cdot}5772,\\
\sigma^{(1)} &= 0{\cdot}926 - 0{\cdot}5184x^{1/3}\left\{1 + \frac{4}{(3x)} - \frac{14}{(3x)^2} + \frac{100}{(3x)^3}\right\},\\
\sigma^{(2)} &= 0{\cdot}134 - 0{\cdot}0744x^{2/3}\left\{1 + \frac{6}{(3x)} - \frac{16}{(3x)^2} + \frac{128}{(3x)^3}\right\},
\end{aligned}\right\}\ x \geqslant 20. \tag{35}$$

Table III gives values of x_1S_1, $x_2S_2/2$ and $\sum_1^\infty x_nS_n/n$ for various values of (T/Z^2). As compared with the results obtained using the asymptotic expansion, the expression

$$\alpha_{\text{tot}} = \mathscr{D}Z\lambda^{1/2}\{0{\cdot}4288 + \tfrac{1}{2}\ln\lambda + 0{\cdot}469\lambda^{-1/3}\} \tag{36}$$

is in error by less than 0·5 per cent for $(T/Z^2) \leqslant 10^5$, by 3 per cent for $(T/Z^2) = 10^6$ and by 31 per cent for $(T/Z^2) = 5\times10^6$ °K.

TABLE III

(T/Z^2) °K	x_1S_1	$\dfrac{x_2S_2}{2}$	$\sum_{n=1}^{\infty}\dfrac{x_nS_n}{n}$
10^3	0·775	0·427	3·046
5×10^3	0·763	0·399	2·305
10^4	0·748	0·371	1·998
5×10^4	0·660	0·260	1·324
10^5	0·585	0·202	1·055
5×10^5	0·350	0·089	0·529
10^6	0·252	0·056	0·364
5×10^6	0·094	0·018	0·127

3. *The energy of electrons recombining on level n.*—Let $kT\beta_nN_eN_+$ be the kinetic energy, per unit volume per unit time, of the electrons recombining on level n. We obtain

$$\beta_n(Z,T) = \mathscr{D}Z\frac{\lambda^{1/2}}{n}x_n^2R_n(\lambda) \tag{37}$$

where

$$R_n(\lambda) = \int_0^\infty \frac{g_{\text{II}}\,e^{-x_nu}}{(1+u)}u\,du. \tag{38}$$

Using (10) we have

$$R_n(\lambda) = R^{(0)}(x_n) + \lambda^{-1/3}R^{(1)}(x_n) + \lambda^{-2/3}R^{(2)}(x_n) \tag{39}$$

TABLE IV

x	$xR^{(1)}$	$xR^{(2)}$	$\rho^{(1)}$	$\rho^{(2)}$
0·02	·139	−·044	·0029	−·0009
·03	·134	−·043	·0043	−·0013
·04	·129	−·043	·0056	−·0018
·05	·124	−·042	·0069	−·0022
·06	·120	−·042	·0081	−·0026
·07	·116	−·042	·0093	−·0031
·08	·112	−·041	·0104	−·0035
·09	·108	−·041	·0115	−·0039
·10	·105	−·041	·0126	−·0043
·12	·099	−·040	·0146	−·0051
·14	·093	−·040	·0165	−·0059
·16	·088	−·039	·0184	−·0067
·18	·084	−·039	·0201	−·0075
·20	·080	−·039	·0217	−·0083
·25	·070	−·038	·0255	−·0102
·3	·062	−·037	·0288	−·0121
·35	·055	−·036	·0317	−·0139
·4	·049	−·035	·0343	−·0157
·5	·038	−·034	·0377	−·0191
·6	·030	−·034	·0411	−·0225
·7	·023	−·033	·0445	−·0259
·8	·018	−·032	·0477	−·0291
·9	·013	−·031	·0509	−·0323
1·0	+·008	−·031	·0540	−·0354

x	$xR^{(1)}$	$xR^{(2)}$	$\rho^{(1)}$	$\rho^{(2)}$
1·0	+·008	−·031	·054	−·035
1·5	−·008	−·029	·054	−·050
2·0	−·013	−·027	·049	−·064
2·5	−·017	−·026	·041	−·078
3·0	−·0204	−·0255	·032	−·091
4	−·0234	−·0240	+·010	−·115
5	−·0245	−·0230	−·014	−·139
6	−·0247	−·0223	−·038	−·161
7	−·0247	−·0216	−·063	−·183
8	−·0244	−·0210	−·088	−·205
9	−·0238	−·0205	−·112	−·225
10	−·0232	−·0199	−·135	−·246
12	−·0221	−·0190	−·181	−·285
14	−·0211	−·0183	−·224	−·322
16	−·0201	−·0177	−·265	−·359
18	−·0191	−·0172	−·304	−·393
20	−·0183	−·0168	−·342	−·427

where

$$xR^{(0)} = 1 - xS^{(0)}, \tag{40}$$

$$xR^{(1)} = -0{\cdot}1728x^{4/3}\frac{d}{dx}\mathrm{X}_1(x), \tag{41}$$

$$xR^{(2)} = 0{\cdot}0496x^{5/3}\frac{d}{dx}\mathrm{X}_2(x). \tag{42}$$

The functions $R^{(1)}$ and $R^{(2)}$ are given in Table IV for $0{\cdot}02 \leqslant x \leqslant 20$. For small x,

$$\left.\begin{aligned} xR^{(1)} &= 0{\cdot}1543(1-8x) + 1{\cdot}944x^{4/3} \\ xR^{(2)} &= -0{\cdot}0448\left(1-\frac{3}{2}x\right) - 0{\cdot}268x^{5/3} \end{aligned}\right\} x \leqslant 0{\cdot}02, \tag{43}$$

and for large x,

$$\left.\begin{aligned} xR^{(1)} &= -0{\cdot}1728x^{-2/3}\left\{1 - \frac{16}{(3x)} + \frac{210}{(3x)^2} - \frac{3200}{(3x)^3} + \frac{57200}{(3x)^4}\right\} \\ xR^{(2)} &= -0{\cdot}0496x^{-1/3}\left\{1 - \frac{6}{(3x)} + \frac{96}{(3x)^2} - \frac{1792}{(3x)^3}\right\} \end{aligned}\right\} x \geqslant 20. \tag{44}$$

For the total kinetic energy loss due to recombination we require

$$\beta_{\mathrm{tot}} = \sum_{n'=1}^{\infty} \beta_{n'}. \tag{45}$$

For large n,

$$\sum_{n'=n}^{\infty} \beta_{n'} \simeq \frac{1}{2}\beta_n + \int_n^{\infty} \beta_{n'}\,dn' = \frac{1}{2}\mathscr{D}Z\lambda^{1/2}\left\{\frac{x_n^{\,2}}{n}R_n(\lambda) + \rho_n(\lambda)\right\} \tag{46}$$

where

$$\rho_n(\lambda) = \rho^{(0)}(x_n) + \lambda^{-1/3}\rho^{(1)}(x_n) + \lambda^{-2/3}\rho^{(2)}(x_n) + \ldots \tag{47}$$

and

$$\rho^{(p)}(x) = \int_0^x x' R^{(p)}(x')\,dx'. \tag{48}$$

We obtain

$$\rho^{(0)}(x) = (1-x)S^{(0)}(x) + \ln x + \gamma = \sigma^{(0)}(x) - xS^{(0)}(x). \tag{49}$$

The functions $\rho^{(1)}$ and $\rho^{(2)}$ are given in Table IV for $0{\cdot}02 \leqslant x \leqslant 20$. For small x,

$$\left.\begin{aligned} \rho^{(1)} &= 0{\cdot}1543x(1-4x) + 0{\cdot}833x^{7/3} \\ \rho^{(2)} &= -0{\cdot}0488x\left(1 - \frac{3}{4}x\right) \end{aligned}\right\} \quad x \leqslant 0{\cdot}02, \tag{50}$$

and for large x,

$$\left.\begin{aligned} \rho^{(1)} &= 1{\cdot}239 - 0{\cdot}5184x^{1/3}\left\{1 + \frac{8}{(3x)} - \frac{42}{(3x)^2} + \frac{400}{(3x)^3}\right\}, \\ \rho^{(2)} &= 0{\cdot}225 - 0{\cdot}0744x^{2/3}\left\{1 + \frac{12}{(3x)} - \frac{48}{(3x)^2} + \frac{512}{(3x)^3}\right\} \end{aligned}\right\} \quad x \geqslant 20. \tag{51}$$

Table V gives values of $x_1^{\,2}R_1$ and $\sum_1^{\infty} x_n^{\,2}R_n/n$ for various values of (T/Z^2). As compared with the results obtained using the asymptotic expansion, the expression

$$\beta_{\text{tot}} = \mathscr{D}Z\lambda^{1/2}\left\{-0{\cdot}0713 + \frac{1}{2}\ln\lambda + 0{\cdot}640\lambda^{-1/3}\right\} \tag{52}$$

is in error by less than 1 per cent for $(T/Z^2) < 5\times10^4$ and by 4 per cent for $(T/Z^2) = 10^5$ °K.

Table V

(T/Z^2) °K	$x_1^{\,2}R_1$	$\sum_{n=1}^{\infty} \frac{x_n^{\,2}R_n}{n}$
10^3	0·772	2·578
5×10^3	0·748	1·857
10^4	0·722	1·564
5×10^4	0·570	0·926
10^5	0·457	0·679

4. *Accuracy of the calculations.*—The asymptotic expansions will be accurate if the third term is much smaller than the sum of the first two terms. For a given temperature the accuracy will improve with increasing n and, for a given n, the accuracy will improve with decreasing temperature.

In Table VI some values of x_nS_n calculated from the asymptotic expansion are compared with the results of more accurate calculations made by Burgess (5); the results of Burgess are exact in the limit of $T\to0$. For $(T/Z^2) \leqslant 2\times10^4$ °K the maximum error in α_1 will be 2·5 per cent and the errors in α_n for $n>1$ will be smaller than 1 per cent.

TABLE VI

T/Z^2) (°K)	n	$x_n S^{(0)}(x_n)$	$\lambda^{-1/3} x_n S^{(1)}(x_n)$	$\lambda^{-2/3} x_n S^{(2)}(x_n)$	$x_n S_n(\lambda)$ (Asymptotic)	$x_n S_n(\lambda)$ (Burgess)	Difference
0	1	1	−·1728	−·0496	·7776	·7973	+·0197
	2	1	−·1089	−·0197	·8714	·8763	+·0048
	3	1	−·0831	−·0115	·9054	·9075	+·0021
10^4	1	·9434	−·1480	−·0470	·748	·763	+·015
	2	·8237	−·0645	−·0169	·742	·748	+·006
	3	·7001	−·0305	−·0089	·661	·660	−·001
$\times 10^4$	1	·8970	−·1293	−·0451	·723	·726	+·003
	2	·7203	−·0436	−·0156	·661	·663	+·002
	3	·5708	−·0148	−·0081	·548	·549	+·001

From the magnitudes of the various terms in the expansions, it is estimated that our results for α_{tot} may be in error by about 10 per cent for $(T/Z^2)=10^6$ and by about 25 per cent for $(T/Z^2)=5\times 10^6$ °K.

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